

SystemC + GCC (MinGW) + CMake Setup Guide on Windows

This document provides a complete flow to: 1. Install GCC/MinGW 2. Build and install SystemC 3. Create and build a SystemC project using CMake 4. Run the executable 5. Debug common issues

1. Prerequisites

Install the following:

- CMake
- MinGW-w64 GCC/G++
- SystemC source package

Verify installation:

```
g++ --version  
cmake --version
```

Expected:

```
g++ (Rev...) 16.x.x  
cmake version 3.x
```

Ensure MinGW bin directory is in PATH:

Example:

```
C:\mingw64\bin
```

Verify:

```
g++ --version
```

2. Build and Install SystemC

Assume:

```
D:\Practive_Learning\SystemC\systemc
```

contains SystemC source.

Create build directory:

```
D:
cd D:\Practive_Learning\SystemC\systemc

mkdir build_gcc
cd build_gcc
```

Configure:

```
cmake .. ^
-G "MinGW Makefiles" ^
-DCMAKE_C_COMPILER=gcc ^
-DCMAKE_CXX_COMPILER=g++ ^
-DCMAKE_CXX_STANDARD=17 ^
-DCMAKE_CXX_STANDARD_REQUIRED=ON ^
-DCMAKE_INSTALL_PREFIX=../install_gcc
```

Build:

```
mingw32-make -j4
```

Install:

```
mingw32-make install
```

After installation:

```
D:\Practive_Learning\SystemC\systemc\install_gcc
```

should contain:

```
include
lib
share
```

Verify:

```
D:\Practive_Learning\SystemC\systemc\install_gcc\lib\libsystemc.a
```

exists.

3. Project Directory Structure

Example:

```
E:\sample
|
├─ main.cpp
├─ i2c_controller.cpp
├─ i2c_controller.h
├─ i2c_registers.cpp
├─ i2c_registers.h
├─ CMakeLists.txt
|
└─ build_gcc
```

Create build directory:

```
mkdir build_gcc
```

4. CMakeLists.txt

Use:

```
cmake_minimum_required(VERSION 3.16)

project(i2c_project LANGUAGES CXX)
```

```
set(CMAKE_CXX_STANDARD 17)
set(CMAKE_CXX_STANDARD_REQUIRED ON)

set(CMAKE_PREFIX_PATH
"D:/Practive_Learning/SystemC/systemc/install_gcc")

find_package(SystemCLanguage CONFIG REQUIRED)

add_executable(i2c_project
    main.cpp
    i2c_controller.cpp
    i2c_registers.cpp
)

target_link_libraries(i2c_project
PRIVATE
SystemC::systemc
)
```

5. Configure Project

Move into build directory:

```
cd E:\sample\build_gcc
```

Configure:

```
cmake .. ^
-G "MinGW Makefiles" ^
-DCMAKE_C_COMPILER=C:/mingw64/bin/gcc.exe ^
-DCMAKE_CXX_COMPILER=C:/mingw64/bin/g++.exe
```

Expected:

```
-- The C compiler identification is GNU
-- The CXX compiler identification is GNU
```

Not:

```
MSVC  
Visual Studio  
cl.exe
```

6. Build Project

```
mingw32-make
```

Verbose build:

```
mingw32-make VERBOSE=1
```

Executable generated:

```
i2c_project.exe
```

7. Run Executable

In Windows CMD:

```
i2c_project.exe
```

or:

```
.\i2c_project.exe
```

Do NOT use:

```
./i2c_project.exe
```

because that is Linux shell syntax.

8. Typical Successful Output

```
SystemC 3.0.2-Accellera  
Constructor called  
SC_METHOD executed at 10 ns
```

9. Common Errors and Fixes

Error:

```
fatal error: systemc.h: No such file or directory
```

Cause:

SystemC include path missing.

Fix:

Use:

```
find_package(SystemCLanguage CONFIG REQUIRED)
```

and:

```
target_link_libraries(  
PRIVATE  
SystemC::systemc)
```

Error:

```
undefined reference to  
sc_api_version_3_0_2_cxx201703L
```

Cause:

C++ ABI mismatch or incorrect linkage.

Fix:

Rebuild SystemC using:

```
-DCMAKE_CXX_STANDARD=17
```

and use:

```
find_package(SystemCLanguage CONFIG REQUIRED)
```

instead of manual linking.

CMake Still Uses MSVC

Delete:

```
CMakeCache.txt  
CMakeFiles
```

Then configure again:

```
cmake .. ^  
-G "MinGW Makefiles" ^  
-DCMAKE_C_COMPILER=gcc ^  
-DCMAKE_CXX_COMPILER=g++
```

Final Working Flow

```
mkdir build_gcc  
cd build_gcc  
  
cmake .. ^  
-G "MinGW Makefiles" ^  
-DCMAKE_C_COMPILER=gcc ^  
-DCMAKE_CXX_COMPILER=g++  
  
mingw32-make
```

`i2c_project.exe`

This flow successfully builds and runs SystemC projects using GCC/MinGW and CMake on Windows.